

CHAPTER 19

PROGRAM DESIGN AND TECHNIQUE FOR PLYOMETRIC TRAINING

Chris A. Bailey, PhD, and Clive Brewer, MSc

The authors would like to acknowledge the significant contributions of David H. Potach and Donald A. Chu to this chapter.

After completing this chapter, you will be able to

- explain the physiology of plyometric exercise;
- identify the phases of the stretch-shortening cycle;
- identify the components of a plyometric training program;
- design a safe and effective plyometric training program;
- prescribe effective plyometric exercise progressions and regressions for a variety of individuals;
- recommend proper equipment for use during plyometric exercise; and
- teach correct execution of lower and upper body plyometric exercises.

Plyometric exercise refers to those activities that enable a muscle to reach maximal force in the shortest possible time. *Plyometric* is a

combination of Greek words that literally means to increase measurement (*plio* = more; *metric* = measure) (67). Practically defined, plyometric exercise is a quick, powerful movement using a prestretch, or countermovement, that involves the stretch-shortening cycle (SSC) (64). Furthermore, plyometric exercises are typically classified as being either ballistic or rebounding. **Ballistic exercises** are those that minimally include contributions from the neuromuscular component of the SSC (e.g., squat jump, medicine ball overhead throw). In contrast, **rebound exercises** are those that require physiomechanical SSC contributions (e.g., drop jumps or repeat countermovement jumps). This classification is important because it determines the specific neuromuscular and mechanical demands of each exercise (37). The purpose of plyometric exercise is to increase the power of subsequent movements by using both the natural elastic components of muscle and tendon and the stretch reflex. To effectively use plyometrics as part of a training program, it is important to understand the mechanics and physiology of plyometric exercise, the principles of plyometric training program design, and the methods of safely and effectively performing specific plyometric exercises.

Plyometric Mechanics and Physiology

Functional movements and athletic success depend on both the proper function of all active muscles and the velocity at which these muscular forces are used. The term used to define this force–velocity relationship is **power**. When properly implemented, plyometric training consistently improves an athlete's ability to produce muscle force and power (34, 59). This enhancement affects the impulse an athlete can generate. Impulse, defined as the integral of force over time, directly influences an athlete's performance by changing momentum. Specifically, an increased impulse leads to greater momentum, which allows athletes to accelerate and decelerate more effectively. These rapid changes in velocity are crucial performance qualities across a wide range of sports. These increases are best explained by two proposed models: mechanical and neurophysiological (64).

Mechanical Model of Plyometric Exercise

In the mechanical model, elastic energy in the musculotendinous components is increased with a rapid stretch and then stored (3, 16, 35). When this movement is immediately followed by a concentric muscle action, the stored elastic energy is released, increasing the total force production (3, 16, 35). Of the mechanical model's many elements, it is the series elastic component (SEC) that is the workhorse of plyometric exercise. While the SEC includes some muscular components, it is the tendons that constitute the majority of the SEC. When the musculotendinous unit is stretched, as in an eccentric muscle action, the SEC acts as a spring and is lengthened; as it lengthens, elastic energy is stored and released with subsequent concentric actions. [Figure 19.1](#) provides a schematic illustration of the components of the SSC.

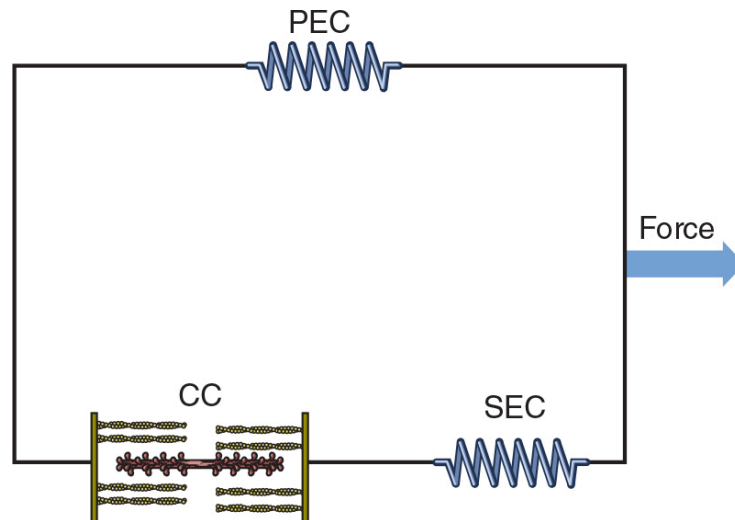


FIGURE 19.1 Mechanical model of skeletal muscle function. The series elastic component (SEC), when stretched, stores elastic energy that increases the force produced. The contractile component (CC) (i.e., actin, myosin, and crossbridges) is the primary source of muscle force during concentric muscle action. The parallel elastic component (PEC) (i.e., epimysium, perimysium, endomysium, and sarcolemma) exerts a passive force with unstimulated muscle stretch.

Adapted from Albert (1995).

Neurophysiological Model of Plyometric Exercise

The neurophysiological model involves the **potentiation** (change in the force–velocity characteristics of the muscle’s contractile components caused by stretch [25]) of the concentric muscle action by use of the stretch reflex (9-12). The **stretch reflex** is the body’s involuntary response to an external stimulus that stretches the muscles (31, 49). This reflexive component of plyometric exercise is primarily composed of muscle spindle activity. **Muscle spindles** are proprioceptive organs that are sensitive to the rate and magnitude of a stretch; when a quick stretch is detected, muscular activity reflexively increases (31, 49). During plyometric exercises, the muscle spindles are stimulated by a rapid stretch, causing a reflexive muscle action. The **series elastic component (SEC)** and **parallel elastic component (PEC)** are two elastic structures that can be stretched within the muscle during plyometric exercises. The SEC includes tendons and muscle fiber crossbridges. The PEC contains

collagenous structures (e.g., fascicular membranes), which provide stability and protection to musculature. This reflexive response potentiates, or increases, the activity in the agonist muscle, thereby increasing the force the muscle produces (9-12, 41). As in the mechanical model, if a concentric muscle action does not immediately follow a stretch (i.e., if there is too long a time between stretch and concentric action or movement over too large a range), the potentiating ability of the stretch reflex is negated.

Enhancements in measurement techniques using force platforms (kinetic data), motion capture (kinematic data), and electromyography (motor unit activation) have demonstrated that the mechanical and neurophysiological models contribute to the increased production of force seen during plyometric exercise (3, 9-12, 16, 35, 41). For example, Jarvis and colleagues (37) have reported that rebound activities elicit enhanced neurophysiological contributions when compared to ballistic activities and that increased mechanical outputs exhibited during high-impact plyometrics are largely the result of mechanical factors associated with potential energy production (i.e., mechanical contributions).

Stretch-Shortening Cycle

The **stretch-shortening cycle (SSC)** employs the energy storage capabilities of the SEC and stimulation of the stretch reflex to facilitate a maximal increase in muscle recruitment over a minimal amount of time. The SSC involves three distinct phases as shown in [table 19.1](#). While the table delineates the SSC's individual mechanical and neurophysiological events during each phase, it is important to remember that all the events listed do not necessarily occur within the given phase. That is, some of the events may last longer or may require less time than allowed in the given phase. Phase I is the **eccentric phase**, which is the time from resting position to the bottom of the countermovement that involves preloading the agonist muscle group or groups. During this phase, the SEC stores elastic energy, and the muscle spindles are stimulated. As the muscle

spindles are stretched, they send a signal to the ventral root of the spinal cord via the type Ia afferent nerve fibers.

TABLE 19.1 Stretch-Shortening Cycle

Phase	Action	Physiological event
I—Eccentric	Stretch of the agonist muscle	Elastic energy is stored in the series elastic component. Muscle spindles are stimulated.
II—Amortization	Pause between phases I and III	Type Ia afferent nerves synapse with alpha motor neurons. Alpha motor neurons transmit signals to agonist muscle group.
III—Concentric	Shortening of agonist muscle fibers	Elastic energy is released from the series elastic component. Alpha motor neurons stimulate the agonist muscle group.

To visualize the eccentric phase, consider a countermovement vertical jump. **Stretch reflex latency** (the time from reflex stimulation to the beginning of agonist muscle activity) largely depends on nerve conduction velocities and therefore increases with greater distances from the spinal cord (i.e., longer nerves) (40, 42, 44, 54). Quadriceps and gastrocnemius stretch reflexes typically range from 20 to 30 milliseconds and from 30 to 45 milliseconds, respectively (40, 53).

Phase II is the time between the eccentric and concentric phases and is termed the **amortization** (or **coupling**) **phase**. This is the time from the end of the eccentric phase to the initiation of the concentric muscle action. There is a delay between the eccentric and concentric muscle actions during which type Ia afferent nerves synapse with the alpha motor neurons in the ventral root of the spinal cord. The alpha motor neurons then transmit signals to the agonist muscle group. This phase of the SSC is perhaps the most crucial in allowing greater power production; its duration must be kept short. If the amortization phase lasts too long, the energy

stored during the eccentric phase dissipates as heat, and the stretch reflex will not increase muscle activity during the upcoming concentric phase (14). Research has shown that increasing time in the amortization phase results in increases in elastic energy lost in the bench press compared to no delay between eccentric and concentric phases (energy losses were 0.35 s = 25%, 0.9 s = 52%, 1 s = 55%, and 1.5 s = 70%) (66). Consider the countermovement vertical jump mentioned earlier. Once the countermovement has stopped, the amortization phase has begun. As soon as movement begins again, the amortization phase has ended.

The **concentric phase**, phase III, is the body's response to the eccentric and amortization phases. In this phase, the energy stored in the SEC during the eccentric phase either is used to increase the force of the subsequent movement or is dissipated as heat. This stored elastic energy increases the force produced during the concentric phase movement beyond that of an isolated concentric muscle action (15, 58). In addition, the alpha motor neurons stimulate the agonist muscle group, resulting in a reflexive concentric muscle action (i.e., the stretch reflex). The efficiency of these subsystems is essential to the proper performance of plyometric exercises. Again, visualize the vertical jump. As soon as movement begins in an upward direction, the amortization phase has ended, and the concentric phase of the SSC has begun. In this example, one of the agonist muscles is the quadriceps (rectus femoris, vastus lateralis, vastus medialis, vastus intermedius). During the countermovement, the quadriceps undergoes a rapid stretch (eccentric phase); there is a delay in movement (amortization phase), and then the muscle concentrically extends the knee, allowing the athlete to push off the ground (concentric phase).

The rate of musculotendinous stretch is vital to plyometric exercise (41). This can be influenced by the type of plyometric activity and the ground contact technique (13). A high stretch rate results in greater muscle recruitment and activity during the SSC concentric phase. The importance of the stretch rate may be

illustrated by three different vertical jump tests: a static squat jump, a countermovement jump, and an approach jump with several steps. As the rate of stretch increases, an athlete's absolute performance in these tests improves; the static squat jump results in the lowest jump height, and the approach jump results in the highest. The static squat jump requires the athlete to get into a squatting position (i.e., 90-degree hip flexion and 90-degree knee flexion) followed by a jump up. This jump does not use stored elastic energy and is too slow to allow potentiation from the stretch reflex because there is essentially no eccentric phase. The countermovement jump uses a rapid eccentric element (i.e., a partial squat) followed immediately by rapid concentric muscle activity (i.e., a jump up). The rapid eccentric phase allows the athlete to store and use elastic energy in the stretched musculotendinous unit and stimulates the stretch reflex, thereby potentiating muscle activity (7, 33). The approach (one- or two-step) jump uses an even quicker, more forceful eccentric phase than the countermovement jump; the increased rate of stretch during the eccentric phase allows a further increase in vertical jump height (4, 6, 8, 29).

The type of ground contact (and, hence, the rate of stretch and duration of the amortization phase, which affects the return of elastic energy) is influenced by the technique used during rebound actions. Cappa and Behm (13) demonstrated that flat-foot landing techniques resulted in significantly longer ground contact times in comparison to active forefoot landings (which also had a 45% greater rate of force production) due to the increased contribution of neuromuscular and elastic force production. Strength and conditioning professionals should note that technique is therefore an essential factor in the effectiveness of the SSC, rate of force development, and subsequent kinematic variables of the movement.

Strength and conditioning professionals may consider that the duration of the amortization phase may lead to activities that promote different movement

characteristics. To this end, Schmidtbleicher (56) categorized plyometric activities as either slow SSC (>0.25 s) or fast SSC (<0.25 s), depending on their ground contact time.

As well as enhancing nonvoluntary neuromuscular outputs, the SSC serves a protective function within the myotendinous structures of the musculoskeletal system. The Golgi tendon organ is encapsulated within the serial components of the myotendinous junction. This is a proprioceptive mechanism that detects the rate and magnitude of force being applied to the respective tendon and will cause an inverse stretch reflex to occur past a tensile threshold, which immediately inhibits the excitatory signal through the motor neuron to the motor unit. Progressive plyometric training serves to desensitize the Golgi tendon organ, thus increasing the excitability potential of the motor unit for increased performance gains.

The SSC combines mechanical and neurophysiological mechanisms and is the basis of plyometric exercise. A rapid eccentric muscle action stimulates the stretch reflex and storage of elastic energy, which results in an increase in the force produced during the subsequent concentric action.

Program Design

Plyometric exercise prescription is similar to resistance and aerobic exercise prescriptions—mode, intensity, frequency, duration, volume, density, technique development and progression, and a warm-up period must all be dynamic variables incorporated in the design of a sound plyometric training program. Unfortunately, there is minimal research demarcating optimal program variables for the design of plyometric exercise programs. Therefore, when prescribing plyometric exercise, strength and conditioning professionals must rely on the available research, practical experience, and the increasing availability of motion capture and force plate technology to guide decision making.

Needs Analysis

To properly design a plyometric training program, the strength and conditioning professional must analyze the needs of the athlete by evaluating his or her sport, sport position, anthropometry, and training status. Each sport and position has its own unique requirements; some requirements are unique because of the movements involved, while others have distinctive injury profiles and risks. The anthropometry of the athlete is important, because heavier athletes face exponentially higher loads in gravity-assisted exercises than athletes with a lower mass; therefore, athletes' body weight and their strength relative to their body weight become important considerations. Further, each athlete possesses a unique training status; some may be new to training and have never performed plyometric exercises, while others may have been injured. Each of these populations of athletes requires a different approach to plyometric training. By understanding each sport's individual requirements, the positions within the sport, and the needs of each athlete, the strength and conditioning professional is better able to design a safe, effective plyometric training program.

Mode

The mode of plyometric training is determined by the body region performing the given exercise. For example, a single-leg hop is a lower body rebounding plyometric exercise, while a two-hand kneeling medicine ball throw is an upper body exercise that is performed ballistically. Modes of plyometric exercise are discussed in the paragraphs that follow.

Lower Body Plyometrics

Lower body plyometrics are appropriate for virtually any athlete and any sport or rehabilitation context. Because most sports require athletes to produce a maximal amount of muscular force in a short amount of time, lower body plyometrics provide an ideal and highly variable training stimulus that positively transfers training benefits to a variety of sports. For example, sprinters require sagittal plane movements, whereas most other sports require combinations of sagittal plane, frontal plane, and occasional transverse plane movements to be successful in competition. Demand can differ within positions in each sport: A middle blocker in volleyball may require predominantly lateral and vertical movements, whereas an outside hitter may have an increased emphasis on horizontal into vertical movements. American football, baseball, soccer, and basketball athletes must make quick, powerful movements and changes of direction in all planes to compete successfully. A basketball center is an example of an athlete who would benefit greatly from a plyometric training program, because the center is required to jump repeatedly for rebounds. To be successful, the center must be able to out-jump the opposing center in order to rebound more loose balls. Conversely, center backs in soccer may benefit from more ballistic actions, because they will typically accelerate into a jump from a running start into a single-foot takeoff.

Lower body plyometric training also gives the ability to produce more force in a shorter amount of time, thereby allowing a higher jump. This should also guide the strength and conditioning

professional to consider that plyometric session design is not limited to actions purely in the sagittal plane, and the transition from one foot to two feet, two feet to one foot, one foot to the other, or one leg to the same leg are often as important for performance enhancement as two-footed actions.

Aerobic endurance athletes can also benefit from participation in a plyometric training program, because repetitive efforts that have neurophysiological and mechanical properties to optimize movement efficiency improve running and cycling economy by allowing muscles to produce more force with less energy.

The various lower body plyometric drills have differing intensity levels and directional movements. The types of lower body plyometric drills include singular and repetitive actions (e.g., **bounds**) using one foot (**hop**) or both feet (**jump**), which may be transitional, in-place, or involve objects such as a box (e.g., a **box jump**) or hurdle. See [table 19.2](#) for a description of these drills.

TABLE 19.2 Lower Body Plyometric Drills

Type	Description
Hops	Takeoff from one foot and landing on the same foot. Volume is either measured by distance traveled or number of foot contacts. This movement develops acceleration and deceleration capabilities in a single limb, with intensity defined by displacement velocity and volume defined by foot contacts. These drills may be singular (ballistic hop) that is in place or horizontal, or reactive (multiple repetitive actions for height or distance).
Bounds	Takeoff from one foot and landing, or transitioning through, to the other foot. Depending on the training status of the athlete (rehab vs. performance), these drills may be singular (ballistic bound) or reactive (multiple repetitive actions with alternating ground contact). Bounds may result in greater horizontal velocity than hops and jumps and therefore are ideal for developing accelerations and decelerations with bilateral weight transfer. Intensity is defined by the horizontal velocity of the action. Volume is typically measured by distance but may be through the number of repetitions performed.
Jumps	Takeoff and landing with two feet that may be horizontal or vertical in nature. The height or length of a jump determines the intensity of the action. Volume is determined by the number of foot contacts. Jumps may be in place, have

Type	Description
	multiple repetitive actions, involve equipment such as boxes or hurdles, or consist of a combination of these attributes.
Depth jumps and depth hops	Depth jumps and hops ("shock" actions) use gravity and the athlete's weight to increase exercise intensity. The athlete assumes a position on a box, steps off, lands, and immediately transitions vertically, horizontally, laterally, or to another box. The height of the box determines the intensity of the action, and the volume is determined by the number of foot contacts. Repetitive jumps onto boxes or over barriers may be added in sequence.

Upper Body Plyometrics

Rapid, powerful upper body movements are requisites for several sports and activities, including baseball, softball, tennis, golf, and throws in track and field (i.e., the shot put, discus, and javelin). For example, an elite baseball pitcher routinely throws a baseball at 80 to 100 miles per hour (129-161 km/h). It is important to understand that to reach velocities of this magnitude, throwing actions have to be initiated with the legs and require hip–shoulder separation and high rotational velocities in the hip and thoracic spine (e.g., a baseball pitcher's shoulder joint must move at more than 6,000 degrees per s) (23, 26, 27, 53). Plyometric training of the shoulder joint may not only increase pitching velocity, but it also may prevent injury to the shoulder and elbow joints. Such upper body actions need to be linked with impulses generated through segmental actions initiated in the lower body if they are to be effective in the transfer to performance.

Plyometric drills for the upper body are not used as often as those for the lower body and have been studied less extensively, but they are nonetheless essential to athletes who require upper body power (52). Plyometrics for the upper body (e.g., **throws** and **slams**) are typically ballistic in nature ([table 19.3](#)).

TABLE 19.3 Upper Body Plyometric Drills

Type	Description
------	-------------

Type	Description
Throws	Single or both arms projecting a medicine ball or other external weight either horizontally or vertically. Progressions include increased mass of the implement, increased target distance, or change of emphasis in complexity of the movement (e.g., kneeling vs. standing throw).
Slams	Generally, a double-arm movement forcefully bringing a medicine ball or other implement directly down to the ground. Progressions include increased mass of the medicine ball or increased target velocity of the medicine ball.
Pull	Ballistically pulling the body in a vertical direction such as an explosive supine row or a pull-up. The position of the center of mass and range of action will be determined by the athlete's strength and movement competence.
Push	Pressing the body away from the ground in a prone position. Progressions need to consider the athlete's relative strength levels and include incorporating shock methodology with varying depth of drop-to-ground contact, decreasing flight time to increase movement velocity, or increasing complexity through lateral or overhead movements.

Note: Although these are classified as upper body movements, some throws and slams are total body in nature. If the upper body is not isolated, many actions are initiated with the hips, knees, and ankles and a postural force-multiplication enables the object to be projected from the upper body. This is entirely appropriate if this is a desired objective within the athlete's program.

Trunk Plyometrics

In general, it is difficult to perform *true* plyometric drills that directly target trunk musculature ([table 19.4](#)) because all the requisite plyometric elements may not be present. As with upper body plyometric exercises, after the eccentric phase of the SSC there is likely some storage of elastic energy during a plyometric (so to speak) trunk drill.

TABLE 19.4 Trunk Plyometric Drills

Type	Description
Lower body anchored	Either suspended in a prone position or in a traditional sit-up position with the feet anchored so that the upper body is free to move. Movements can be progressed by incorporating or increasing an external resistance or movement velocity.

Type	Description
Upper body anchored	Lying supine, holding a fixed object or a partner's legs, while the lower body is free to move. Movements can be progressed by incorporating or increasing external resistance, movement velocity, or plane of motion (e.g., increased transverse plane involvement is more complex than sagittal plane only).
Throws	Forcefully projecting a medicine ball or other external weight in a specified direction. The mass of the implement will influence the resultant movement velocity and should be considered in respect to the mass-velocity-momentum equation and the force production capability of the athlete.

Exercises for the trunk may be performed ballistically provided that movement modifications are made. Specifically, the exercise movement must be shorter and quicker to allow stimulation and use of the stretch reflex. The relatively large range of motion and the time needed to complete the movement do not permit reflexive potentiation of the abdominal muscles. The exercises can be modified to decrease both the range of motion and time, thereby allowing the agonist muscles to be potentiated.

Intensity

Plyometric intensity refers to the stress placed on muscles, connective tissues, and joints during an exercise. It is primarily determined by the type of drill performed, technique used, and arrangement of equipment. Additionally, it is important to consider the complexity of the movement when assessing intensity. For example, multiple hurdle hops in a straight line can involve higher linear speeds at a relatively lower intensity. However, if the same hurdles are arranged in a square that requires the athlete to change direction with each hop, the intensity will be significantly higher. This increase in intensity occurs because athletes must decelerate their momentum before each direction change. They must then use the plyometric response to propel themselves in a new direction. As a result, both the intensity of the drill and the ground contact time can increase substantially when directional changes are incorporated.

The intensity of plyometric drills is, therefore, multifactorial and should be considered on a continuum from low to high intensity depending on a number of planning factors. For example, skips are considered a relatively low-intensity drill, while **depth jumps** are considered a higher intensity drill because they place a higher stress on the muscles and joints ([figure 19.2](#)).

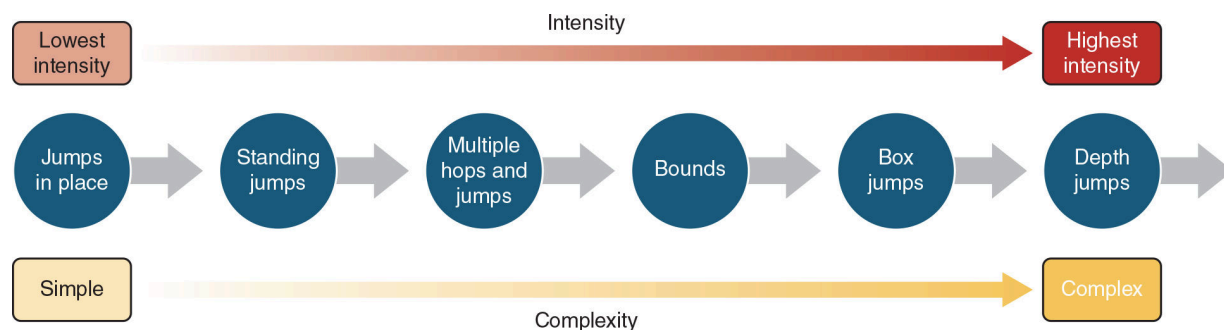


FIGURE 19.2 Continuum of plyometric exercises.

In addition to the type of drill, several other factors also affect plyometric intensity ([table 19.5](#)). Generally, as intensity increases, volume should decrease (57). Because the intensity of plyometric exercise can vary significantly, careful consideration must be given to choosing proper drills during a specific training cycle.

TABLE 19.5 Factors Affecting the Prescribed Intensity of Lower Body Plyometric Drills

Factor	Effect
Points of contact	The ground reaction force during single-leg lower body plyometric drills places more stress on an extremity's muscles, connective tissues, and joints than during double-leg plyometric drills.
Speed	Greater horizontal velocity increases the intensity of the drill.
Technique	Ankle, knee, and hip stiffness and the ability to maintain postural alignment influence ground contact time, ground reactional forces, eccentric muscle actions, and injury risk.
Strength	Higher levels of strength-to-body weight reduce the relative intensity of a drill where mass is accelerated toward the ground by gravity (i.e., any

Factor	Effect
	form of reactive plyometric drill).
Directional change	Changing direction requires deceleration and redirected acceleration of the athlete's center of mass. This increases the impulse required in the action and typically requires increased ground contact time.
Vertical component of the drill	The higher the body's center of gravity, the greater the force on landing.
Mass	The greater the athlete's body weight, the more stress is placed on muscles, connective tissues, and joints. External load in the form of bars, weight vests, ankle weights, and wrist weights can be added to the body to increase a drill's intensity.

Much like exercise selection for resistance training, specific consideration should be given to the muscle actions that are incorporated into each plyometric exercise because this will alter the muscle activation and resulting intensity. Reactive exercises require athletes to reduce forces in a landing impact prior to propelling themselves in the desired direction, while ballistic exercises may not (37). For example, a single countermovement jump does not require an athlete to absorb the same force as a depth jump. This realization is important, because research indicates that muscle activation, power, and relative impulse are all higher in rebound exercises compared to ballistic exercises, which demonstrates an increased level of intensity (37).

When considering exercise intensity, it is important to take note of the exercise intensity overall as well as at the joint level. Research has shown that intensity is joint specific when considering the hips, knees, and ankles of several exercises (61). For example, the intensity of a backward jump is higher at the hips and ankles than it is at the knees. Exercise modifications based on the joint considerations of an athlete are not uncommon, and considering the intensity applied at each joint that is involved in an exercise will likely result in more effective programming and adaptation.

Frequency

Frequency refers to the number of plyometric training sessions per week, typically ranging from one to four. This frequency depends on the athletes' sport, their experience with plyometric training, their overall training status, the volume and intensity of plyometrics within the broader resistance training program, and the time of year (e.g., off-season or in-season). For example, during the season, one session per week is appropriate for American football players, while two or three sessions per week are common for track and field athletes (2). During the off-season, plyometric training frequency may increase to two or three sessions per week for American football players and to three or four sessions per week for track and field athletes (2).

As with other program variables, research on the optimal frequency for plyometric training is limited. Because the literature is sparse, strength and conditioning professionals often rely on practical experience when determining the frequency with which athletes train using plyometric exercise. Rather than concentrating on the frequency, many authors suggest relying more on the recovery time between plyometric training sessions (19). Forty-eight to 72 hours between plyometric sessions is a typical recovery time guideline for prescribing plyometrics (19); using these typical recovery times, athletes commonly perform two to three plyometric sessions per week. As with resistance training, recovery between workouts must be adequate to prevent overtraining (depending on the sport, training status, and training intensity). Furthermore, moderate to high volumes for a given body area should not be performed on successive days (51).

Recovery

Because plyometric drills involve maximal efforts to improve power, they are typically fatigue inducing and require complete and adequate recovery (i.e., sufficient time between repetitions, sets, and workouts) (51). Recovery for depth jumps may consist of 5 to

10 seconds of rest between repetitions and 2 to 5 minutes between sets. The time between sets is determined by a proper work-to-rest ratio (i.e., 1:5 to 1:10) and is specific to the volume and type of drill being performed. Drills should not be thought of as aerobic endurance training but as power training. Plyometric training can induce fatigue and inflammatory responses that can significantly increase delayed onset of muscular soreness and may impair jumping performance for up to 72 hours after a session. While these effects are noticeable in power-related measurements, strength levels may remain unchanged (5, 18, 60). Given this prolonged recovery period, strength and conditioning professionals should carefully schedule plyometric sessions to allow adequate recovery time between workouts and before competitions. This approach helps ensure optimal performance and reduces the risk of overtraining or injury.

Volume

Plyometric volume is typically expressed as the number of repetitions and sets performed during a given training session ([table 19.6](#)). Lower body plyometric volume is normally given as the number of **foot contacts** (each time a foot, or the feet together, contact the surface) per workout (2, 19) but may also be expressed as distance, as with plyometric bounding. For example, an athlete beginning a plyometric training program may start with a bound for 30 yards (or meters) per repetition but may progress to 100 yards (or meters) per repetition for the same drill. Recommended lower body plyometric volumes vary for athletes of different levels of experience and also with the intensity level of the prescribed exercise. Upper body plyometric volume is typically expressed as the number of throws or catches per workout, and the ballistic nature of these actions typically makes the training volumes reflect those of upper body resistance exercises (i.e., 3-5 sets of 3-5 repetitions).

TABLE 19.6 Plyometric Volumes

Plyometric experience level	Intensity level of the plyometric drill		
	Low	Moderate	High
Beginner (no experience)	80 to 100	Nr to 40	Nr
Intermediate (some experience)	100 to 120	20 to 100	Nr to 20
Advanced (considerable experience)	120 to 200	20 to 120	20 to 40

Note: Volume is given in contacts per session. Nr = not recommended.

Progression

Plyometrics is a form of resistance training and thus must follow the principles of **progressive overload**, which is the systematic increase in training frequency, volume, and intensity in various combinations. In simplistic models, as intensity increases, volume decreases. This relationship, however, is variable depending on the athlete's training status and the nature of the activity prescribed. For example, athlete A who has minimal plyometric experience and a low training status may do 3 sets of 10 repetitions of the pogo jump to specifically train the SSC in the gastrocnemius–soleus complex, but athlete B, who has a higher training age, may do the same drill with an emphasis on minimizing ground contact time, and elite athlete C may do the drill at the end of a warm-up to simply potentiate the neuromuscular system. Further, after the pogo jump, athlete A may do jumps over short hurdles, but athlete B may do the depth jump instead and perform 2 sets of 4 repetitions with a 24-inch (61 cm) box. Elite athlete C may also do the depth jump but at a higher volume and intensity, such as 4 sets of 8 repetitions using a 45-inch (114 cm) box.

Other factors such as the athlete's sport, sport season, and training phase can influence the plyometric training schedule, design of a plyometric training session, and the method of progressive overload. An off-season plyometric program for American football, for example, may be performed twice a week, but freshmen versus senior players will do different plyometric activities. Additionally, the plyometric programs performed by skill (position) players versus

offensive or defensive line players may be different. Thus, careful consideration should be given to each athlete's capacity and training status when progressing a plyometric program, because volume and intensity are not always directly related.

Effective plyometric programs include the same variables that are essential to any training program design: mode, intensity, frequency, recovery, volume, and progression.

Program Length

Research has yet to determine an optimal plyometric training program length. Currently, most programs range from 6 to 10 weeks (2, 34); however, vertical jump height improves as quickly as four weeks after the start of a plyometric training program (54). In general, plyometric training should be prescribed similarly to resistance training within a periodized plan. For those sports requiring quick, powerful movements, it is beneficial to perform plyometric exercises throughout the training cycle. The intensity and volume of the selected drills should vary with the training status of the athlete and the sport season.

Warm-Up

As in any training program, a plyometric exercise session should begin with a specific warm-up that incorporates activation, mobilization, and potentiation of the neuromuscular and musculoskeletal system (refer to [chapter 15](#) for movement preparation guidelines). The specific warm-up for plyometric training should consist of low-intensity, dynamic movements that increase in intensity as the warm-up progresses. [Table 19.7](#) provides a list and explanation of examples of specific warm-up drills for plyometric training.

Steps for Implementing a Plyometric Training Program

1. Evaluate the athlete, including the athlete's sport and training history.
2. Establish sport-, position-, and athlete-specific goals.
3. Teach the athlete proper jumping, landing, and throwing technique.
4. Assign proper plyometric training program design variables, addressing intensity, frequency, recovery, and volume.
5. Properly progress the plyometric training program.

Age Considerations

It is becoming more common for younger and older individuals to want to augment the training programs for their sport with plyometric exercise. When these exercises are applied appropriately, these populations can experience the same positive outcomes as other age groups, with minimal risk of injury, with appropriate consideration of prescription to technical competence and strength levels.

Adolescents

Although plyometrics commonly have been viewed as appropriate only for conditioning elite adult athletes, prepubescent and adolescent children may also benefit from training with plyometric and plyometric-like exercises. Besides providing the well-documented muscular power and bone strength adaptations, regular participation in an appropriately designed plyometric training program can better prepare young athletes for the demands of sport practice and competition (20) by enhancing neuromuscular control and performance. Research has yet to determine a universal age at which people are physically able to begin participating in a plyometric training program. An analysis of the body's development provides some insight into the issue. Because the epiphyseal plates of the bones of prepubescent children have yet to close (39, 47), depth jumps and other high-intensity lower body drills are contraindicated (2, 36, 46). Further, and as with all forms of exercise, boys and girls should have the emotional maturity to accept and follow directions and should be able to appreciate the benefits and concerns associated with this method of training. Empirically, 7- and 8-year-olds have been trained in progressive plyometric programs and without detriment to future sport involvement (20).

TABLE 19.7 Examples of Plyometric Warm-Up Drills

Drill type	Explanation
Marching	Mimics running movements Emphasizes posture and movement technique Reinforces proper lower body movements for running
Ground contact	Prepares the body for impact and high-intensity drills: • Rapid pogo jump—not allowing the heel to touch the ground (emphasizes ankle dorsiflexion with active front foot ground contact) • Straight-leg skip or bound—emphasizes negative foot speed (the rapid backward acceleration of the foot) into the ground, minimizes knee flexion, and maximizes hip extension to prepare for the impact of plyometric drills Exaggerated form of reciprocal upper and lower extremity movements Emphasizes quick takeoff and landing Mimics plyometric activities
Footwork	Targets multiplanar acceleration and deceleration mechanics Prepares for changes of direction during plyometric drills Examples: shuttle, shuffle, pattern, and stride drills
Neuromuscular acceleration and deceleration	Multijoint explosive directional changes with full-range movements: • Forward, side, and backward lunge (similar to the forward [step] lunge exercise but multidirectional) • Squat patterns (e.g., a tall reach into a deep snap-down then a squat jump to a standing landing)

Note: These examples are specific preparatory drills that can be incorporated into the RAMP phases of a warm-up for plyometrics; see [chapter 15](#) for more detail.

Plyometric exercise programs for children should be used to develop the neuromuscular control and the anaerobic skills that will carry over to safer participation in sport and athletics, both during childhood and as they advance to higher levels of competition. For example, several research studies cite the benefits of using proper landing technique as a method of reducing an athlete's risk of lower extremity injury (see the Safety Considerations section starting on [page 576](#) for details).

It is extremely important that plyometric exercise programs for children gradually progress from relatively simple drills to more complex drills. It is important to focus on the quality of the movements (e.g., proper body alignment and speed of movement) to develop techniques that will be essential for more advanced exercises. The current evidence suggests that a twice-weekly program for 8 to 10 weeks beginning at 50 to 60 jumps a session and increasing exercise load weekly results in the largest changes in running and jumping performance in young children (38). An alternative program for children who do not have the capability or tolerance for a twice-weekly program would be a low-intensity program for a longer duration (38).

Under proper supervision and with an appropriate program, prepubescent and adolescent children may perform plyometric exercises. Depth jumps and high-intensity lower body plyometrics are contraindicated for this population.

Masters

Masters athletes find that they can maintain their physical capabilities late into life and are looking for additional training insights. When designing a plyometric training program for a masters athlete, the strength and conditioning professional needs to be specific in deciding on the goal of the program. Some primary issues to consider are strength levels (i.e., ensuring that age-related sarcopenia has not degraded muscle function) and any preexisting orthopedic conditions (e.g., osteoarthritis or any sort of surgical joint intervention) or joint degeneration. These call for even greater caution and a more careful use of plyometric exercise. For example, a healthy masters athlete without surgical history who wants to improve his or her running performance should use depth jumps and single-leg exercises cautiously, so alternate-leg bounding and the double-leg hopping would be better choices. Similarly, a masters

runner with a history of knee surgery such as partial meniscus removal or with significant joint degeneration should regard depth jumps and single-leg plyometric exercises as contraindicated and use other forms of plyometrics cautiously.

After consideration of the predispositions related to the masters athlete's physical condition, a plyometric program should be designed according to the same guidelines as outlined for adult athletes, with the following emphases:

- Intensity and volume should be carefully considered relative to the performance capacities of the individual.
- Cautions regarding decreased strength capacities, potential for degenerative joint conditions, and decreased recovery capacities associated with aging should be applied.
- It is important to note how the masters athlete feels after training; soreness may occur, but the program should be modified if chronic or excessive pain or discomfort is present.

Plyometrics and Other Forms of Exercise

Plyometric exercise is only one part of an athlete's overall training program. Many sports and activities use multiple energy systems or require other forms of exercise to properly prepare athletes for competition. Each energy system and sport-specific need must be included in a well-designed training program.

Plyometric Exercise and Resistance Training

A combination of plyometrics and resistance training during a training cycle should be structured to allow maximal efficiency and physical improvement. The following list and [table 19.8](#) provide guidelines for developing a combined program:

- Combine lower body resistance training with upper body plyometrics, and upper body resistance training with lower body plyometrics.
- Performing heavy resistance training sessions and plyometric sessions on the same day is not usually recommended (17, 24). However, many athletes may benefit from **complex training**, a combination of resistance training and plyometric training within the same training session to enhance fast and slow force expression.
- If athletes perform this type of training, adequate recovery is needed between plyometrics and other high-intensity training.
- Plyometrics can comprise a stand-alone training session or be incorporated into a resistance training session. Incorporating plyometrics into a resistance training session may have neuromuscular recruitment benefits through facilitating postactivation potentiation during biomechanically similar movements (22).

TABLE 19.8 Sample Schedule for Integrating Resistance Training and Plyometrics

Day	Resistance training session	Plyometric session	Single session complex*
Monday	Low-intensity (light) upper body	High-intensity (reactive) lower body	High-intensity (heavy) lower body resistance training + Lower body plyometrics
Tuesday	High-intensity (heavy) lower body	Low-intensity upper body	Low-intensity (light) upper body resistance training + Upper body plyometrics
Thursday	High-intensity (heavy) upper body	Low-intensity (ballistic) lower body	High-intensity (heavy) upper body resistance training + Upper body plyometrics
Friday	Low-intensity (light) lower body	High-intensity upper body	Low-intensity (light) lower body resistance training + Lower body plyometrics

*Emphasis on resistance training + plyometrics within a total body session.

The following are classifications and examples of complex training based on Cormier and colleagues (22):

- **Contrast training** involves alternating high-load exercises and high-velocity exercises in contrast pairs in a set-by-set fashion with sufficient inter-set rest to avoid fatigue-induced interference. An example is an American football lineman alternating sets of heavy bench presses with sets of clap push-ups.
- **Ascending complex training** involves sets of a high-velocity exercise followed by sets of a contrasting high-force exercise. An example is a weightlifter doing 2 sets of maximal-effort box jumps before performing mid-thigh power cleans.
- **Descending sets** require multiple high-force exercises to be completed before the execution of high-velocity exercises. An example is a thrower doing 3 sets of 2 repetitions of heavy back squats followed by 3 sets of 4 repetitions of overhead medicine ball throws.